

|                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                      |
|----------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|
| <b>Type: GE4 / OE4 (Generic / Open Elective)</b><br><b>B.Sc(ECS)-II (Semester IV)</b><br><b>Course Title: Computer Networks</b><br><b>(Paper Code: )</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                      |
| <b>Credits: Theory – (2)</b>                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                      |
| <b>Total Lectures:</b> 30 Hrs.                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>Contact Hrs. (L) : 2</b>          |
| <b>University Evaluation:</b> 30 Marks                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>Internal Evaluation:</b> 20 Marks |
| <b>Course Outcomes:</b>                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                      |
| 1.                                                                                                                                                       | Understand the fundamental components of data communication systems, including sender, receiver, transmission medium, message, and protocol.                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                      |
| 2.                                                                                                                                                       | Describe the concept of computer networks, their evolution, and the various uses of networks in modern society.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                      |
| 3.                                                                                                                                                       | Identify key networking components such as hubs, switches, repeaters, bridges, routers, and gateways and understand their respective roles.                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                      |
| 4.                                                                                                                                                       | Compare various network topologies (bus, star, ring, mesh, hybrid) and types of networks (LAN, MAN, WAN).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                      |
| 5.                                                                                                                                                       | Understand inter-networking concepts and how different networks communicate using protocols.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                      |
| 6.                                                                                                                                                       | Explore various applications of the internet, such as email, web browsing, e-commerce, and cloud computing.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                      |
| <b>Unit</b>                                                                                                                                              | <b>Content</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>Lect.</b>                         |
| <b>I</b>                                                                                                                                                 | <b>Introduction to Data Communication and Networking:</b> Data Communication: Components, Data Representation, Data Flow Communication Model Computer N/W: Introduction of Network, Uses of computer network N/W Components: Hubs, Switches, Repeater, Bridges, Routers, Gateways. N/W Topologies, Types of Networks, Inter-networking, Applications of Internet.<br><b>Network Models:</b> Protocols & Standards, Protocol Hierarchies, Design Issues of Layers, Services Primitives, Connection oriented and connection less services Reference Model: ISO-OSI reference model, TCP/IP reference model.              | <b>10</b>                            |
| <b>II</b>                                                                                                                                                | <b>Physical layer:</b> Signals-Analog and Digital Signals, Period, Frequency, Phase, Amplitude, Bandwidth, Bit Rate, Bit Length, Fourier analysis. Transmission Impairment-Attenuation, Distortion, Noise, Transmission Media-Guided Media-Magnetic Media, Twisted Pair, Coaxial Cable, Fiber Optic Cable, Unguided Media- Wireless Radio Waves, Microwaves, Infrared, Satellite Communication Analog Transmission-Modem, Digital Transmission-Pulse Code Modulation, Manchester and Differential Manchester Coding. Modulation and types- Amplitude, Frequency, Phase Transmission Mode-Parallel, Serial, Synchronous | <b>20</b>                            |

|                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |  |
|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
|                         | <p>Transmission, Asynchronous Transmission. Multiplexing and types- Frequency, Time, Wavelength, Switching and types- Circuit, Message, Packet.</p> <p><b>Data link layer:</b> Data link layer Design issues, Error Detection &amp; CorrectionTypes of Errors, Hamming Distance, Error Detection-Parity Check, Cyclic Redundancy Check, Checksum Check Error correction, Data Link ControlFraming, Flow &amp; Error Control, Protocols-Simplex, Stop and Wait, Stop and Wait ARQ, Go Back N ARQ, Selective repeat ARQ. Multiple Access Protocol-ALOHA, CSMA, CSMA/CD, CSMA/CA Channelization, FDMA, TDMA, CDMA</p> |  |
| <b>Reference Books:</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |  |
| 1.                      | Computer Networking by Tannenbaum.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |  |
| 2.                      | Data communication and networking by B A Forouzan                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |  |
| 3.                      | Computer Networks: A Systems Approach by Larry L. Peterson, Bruce S. Davie.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |  |